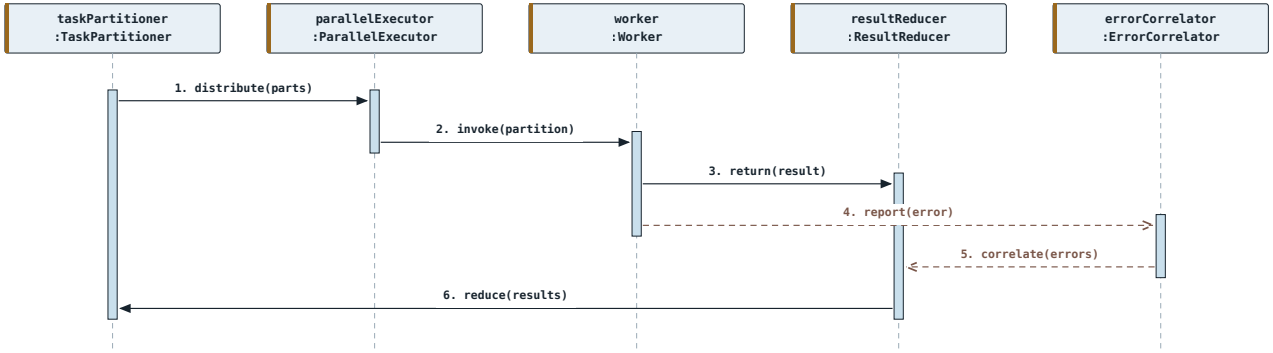
W03 Parallelization

Workflow / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Run independent work or alternative judgments concurrently.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE	
CONTEXT	Independent tasks or alternative judgments can run concurrently.
TRIGGER	A coordinator fans work out and later aggregates it.
STRUCTURE	Partition contract, worker identity, result correlation, and reducer.
EFFECT	All selected results are reconciled into a declared aggregate or failure.

CONTRACT / REQUIRED BEHAVIOR	
REQUIRES	Work partitions are independent enough for concurrency.
ENSURES	Aggregation includes every expected result or records its absence.
INVARIANT	Partial completion is never represented as full agreement.
VIOLATION	Failures, timeouts, and cancellation follow declared quorum or terminal policy.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.